

On the Computational Complexity and Approximation of the General Routing Problem in Euclidean Graphs

AN JUNHYUN

2025-04-24

Abstract

In this note, we analyze a simple upper bound on the additional cost incurred in the General Routing Problem (GRP) when the set of required edges is disconnected. We show that the GRP for Euclidean graphs is NP-hard. We also show that augmenting the required edge set by connecting its components using minimal intercomponent distances yields a connected graph on which the Chinese Postman Problem (CPP) can be solved with a total augmentation cost that is at most triple the optimal augmentation cost of the original GRP instance, and at most twice for two connected component. This result is applicable in general metric graphs, including Euclidean geometric graphs.

1 Introduction

The General Routing Problem (GRP) is a generalization of the Chinese Postman Problem (CPP), in which a subset of edges $R \subseteq E$ must be traversed in a minimum-cost closed walk. When where R is disconnected, it is necessary to add additional edges to ensure the existence of an Eulerian tour. In this work, we provide a simple upper bound on the cost of this augmentation.

2 Preliminaries

Let $G = (V, E)$ be a perfect Euclidean graph, where $V \subset \mathbb{R}^2$ and $E = \binom{V}{2}$. Let $Q \subseteq V$ and $R \subseteq E$ each denote the required vertex and edge set. Suppose the subgraph $G_R = (Q, R)$ contains $k \geq 1$ connected components (consider that the disjoint components have no edges overlapping):

$$G_R = C_1 \cup C_2 \cup \cdots \cup C_k.$$

Define:

$c(e)$: the cost of edge $e \in E$.

For $e = (v_1, v_2)$, $c(e) = d(v_1, v_2) = \|v_1 - v_2\|_2$

$c(R)$: the total cost of edges in R , $c(R) = \sum_{e \in R} c(e)$.

$l(R)$: the minimum additional cost required to augment R into a closed Eulerian walk (i.e., the difference between the cost of an optimal GRP tour and $c(R)$).

H : a minimum-cost set of edges connecting all components of G_R (e.g., a minimum spanning tree over components).

M : a minimum-cost set of edges making all odd vertex of G_R have even degree (e.g., a minimal matching over Euclidean perfect graph).

$J(R)$: the cost of a closed Eulerian walk obtained by an approximation algorithm, specifically $J(R) = c(H' \cup M)$, where H' doubles all edges in H .

3 Main Result

3.1 NP-hardness

Let G , R , Q , and M be as defined above. We show that the Euclidean GRP is NP-hard by reducing from the Euclidean Traveling Salesman Problem (TSP).

Theorem 1. *The General Routing Problem in the Euclidean plane with L_2 distances is NP-hard.*

pf. Consider the special case of the GRP where $R = \emptyset$ and the input graph is a complete graph on a set of points in the Euclidean plane. Then the GRP reduces to finding a shortest Hamiltonian cycle where edges are allowed to appear twice.

Let $T^* = \bigcup_{i=1}^n \{(v_{i-1}, v_i)\}$ be an optimal solution corresponding to the shortest Hamiltonian cycle $(v_0, v_1, \dots, v_n = v_0)$. If some vertex v appears twice as $v_i = v_j = v$ for $i, j < n$. There exists adjacent edge $e_1 = (u, v)$, $e_2 = (v, w)$ so that $e_1, e_2 \in T^*$. Since the edges have cost as norm,

$$c(\{e_1, e_2\}) = c(e_1) + c(e_2) = \|u - v\| + \|v - w\| \geq \|u - w\|$$

By replacing e_1, e_2 with $e = (u, w)$, new solution be $T = T^* \setminus \{e_1, e_2\} \cup \{e\}$. Thus,

$$c(T) = c(T^*) - c(\{e_1, e_2\}) + c(e) \leq c(T^*)$$

T corresponds to cycle that still pass through v , since v appears twice in Hamiltonian cycle. T forms a Hamiltonian cycle with shorter cost than the optimal solution T^* . By contradiction, optimal solution passes through all vertex exactly one time. The optimal solution is solution of Traveling Salesman Problem (TSP) on Euclidean plane.

The Euclidean GRP reduces to the Euclidean TSP, which is known to be NP-complete [1]. Therefore, the Euclidean GRP is at least as hard as Euclidean TSP, and is thus NP-hard. $\square$

Also, for nonempty R , or moreover G_R without trivial graph as connected component, it is NP-hard. We similarly show this by constructing G_R with no trivial graph as connected component, Euclidean GRP for G_R is reduced to Euclidean TSP.

Theorem 2. *The General Routing Problem in the Euclidean plane with nonempty R is also NP-hard.*

pf. Let $P = \{p_i\}_{i=1}^n \subseteq \mathbb{R}^2$ be arbitrary. For finite P , find $s = \max_{i,j \in [1,n] \cap \mathbb{N}} \|p_i - p_j\|$. Let's construct a disjoint graph C_i s as following.

$$C_i = (Q_i, R_i) = (\{p_i, p_{i1}, p_{i2}\}, \{(p_i, p_{i1}), (p_i, p_{i2}), (p_{i1}, p_{i2})\})$$

, where

$$d(p_i, p_{i1}) = d(p_i, p_{i2}) = i \cdot (3s)$$

For any point p_{i1} (or p_{i2}) find a lower bound of distance between disjoint group C_j .

$$d(p_{i1}, p_j) \geq d(p_{i1}, p_i) - d(p_i, p_j) \geq 3is - s = (3i - 1)s \geq 2s$$

Also, for distance between p_{i1} (or p_{i2}) and p_{j1} (or p_{j2}) find a lower bound. WLOG let $i > j$

$$d(p_{i1}, p_{j1}) \geq d(p_{i1}, p_i) - d(p_i, p_{j1}) \geq d(p_{i1}, p_i) - (d(p_i, p_j) + d(p_j, p_{j1})) \geq 3is - s - 3js = (3i - 3j - 1) \cdot s$$

Since $3i - 3j - 1 = 3(i - j) - 1 \geq 2$ for $i > j$, the thus the lower bound is $d(p_{i1}, p_{j1}) \geq 2s$. Therefore, $(\forall p \in Q_j) \quad d(p_{i1}, p) \geq 2s$.

Solve GRP for unions of C_i . Let $Q = \bigcup_{i=1}^n Q_i$, then take Euclidean perfect graph $G = (Q, \binom{Q}{2})$, and $R = \bigcup_{i=1}^n R_i$, then $G_R = (Q, R)$.

Each connected components C_i has even degrees for all vertex. For an optimal solution T^* , in general T^* must contain some set H^* that connects C_i s, and M^* that connects every odd degree vertices to ensure Eulerian connectivity. However, in this case, M^* is not needed, so $T^* = H^*$.

For any connection between C_i, C_j , we've shown the lower bound of point p_{i1} (and p_{i2}). Thus,

$$(\forall p \in Q_j) \quad d(p_{i1}, p) \geq 2s \geq d(p_i, p)$$

Taking edge connecting C_i, C_j , it is always optimal to choose p_i for vertex in C_i . Also, for C_j , taking p_j is optimal. Therefore $M^* \subset \binom{P}{2}$. Therefore, solving T^* is solving TSP in Euclidean space for point P given in complete graph $G_P = (P, \binom{P}{2})$. Therefore, even for nonempty R Euclidean GRP is at least as hard as Euclidean TSP, and is thus NP-hard. $\square$

3.2 Approximation Algorithm

Let G, R, Q , and M be as defined above. Then,

$$J(R) - c(R) \leq 3l(R).$$

That is, the additional cost of solving CPP on $R \cup M$ is at most triple the additional cost in the optimal GRP solution.

pf. Let T^* be an optimal solution to the GRP instance on R . Then T^* must contain some set $H^* \subseteq E \setminus R$ that connects all disjoint components of G_R , and $M^* \subseteq E \setminus R$ that connects every odd degree vertices to ensure Eulerian connectivity. Take all odd vertices of $G_R = (Q, R)$ so that the minimal matching M making all vertex degree even [2]. There is a polynomial time algorithm of finding M in Euclidean space [3] [4]. Since M is a minimum-cost set of edges that connects all components, we have:

$$\text{cost}(M) \leq \text{cost}(M^*) \leq l(R).$$

Next, take minimum distance between disjoint C_i s. Let G_c be a complete graph $G_c = (\mathcal{C}, \binom{\mathcal{C}}{2})$, where $\mathcal{C} = \{C_i | i \leq k\}$. The cost of edge (C_i, C_j) be minimum distance between vertex in C_i and C_j .

Consider for case of $k = 2$.

Since adding T^* forms a Eulerian cycle crossing both C_1 and C_2 , we can give a direction. Take any vertex (WLOG in C_1) be starting point of the cycle. Following the (directed) Eulerian cycle, there exists a edge from $C_1 \rightarrow C_2$ and $C_2 \rightarrow C_1$, since Eulerian cycle contains a vertex in C_2 and end at starting point in C_1 . The edge added between two component for H^* is at least two. The minimum cost of (C_1, C_2) always smaller or equal to any of those edges. Let H'_2 be set of double edge for (C_1, C_2) , typically corresponds to some $v_1 \in C_1$ and $v_2 \in C_2$ s.t. $c(v_1, v_2) = c(C_1, C_2)$. Thus,

$$c(H'_2) = 2c(v_1, v_2) = 2c(C_1, C_2) \leq c(H^*) \leq l(R)$$

For general $k > 2$, we can H in more general way.

A minimum spanning tree (MST) provides a polynomial-time method to connect all components of G_c with the smallest total cost [5] [6] [7]. Let H be the union of edges obtained by MST.

Generate H' by changing all edge in H into double edge. Adding double edge increases vertex degree of 2, so doesn't change the existence of Eulerian cycle. The additional edges $M \cup H'$ for G_R satisfy:

1. H' connect the component C_i s
2. M guarantees the all vertex be even degree

Therefore, $G' = (Q, R \cup M \cup H')$ is connected graph with Eulerian cycle.

Since T^* has H^* that connects every component C_i s, let L be set of all linked C_i, C_j by H^*

$$L = \{(i, j) | u \in C_i, v \in C_j \text{ for } (u, v) \in H^*\}$$

Then $\forall (u, v) \in H^*$, for $u \in C_i, v \in C_j$ $c((u, v)) = d(u, v) \geq d(C_i, C_j)$ by definition of $d(C_i, C_j)$. The total cost of H^* is bounded below

$$c(H^*) \geq \sum_{(i,j) \in L} c(C_i, C_j) \geq c(H)$$

The $c(H)$ is minimum cost for edges that connects all C_i, C_j , so one case of connecting $(i, j) \in L$, the cost of all connected edges are always larger than $c(H)$. Thus,

$$c(H') = 2c(H) \leq 2c(H^*) \leq 2l(R)$$

It follows that the matching cost $J(R) = c(R \cup M \cup H')$ has upper bound as following.

$$J(R) - c(R) = c(M) + c(H') \leq l(R) + 2l(R) = 3l(R).$$

□

4 Discussion

This bound shows that a simple heuristic which first connects components using minimum intercomponent distances and then solves the Chinese Postman Problem yields a tour whose cost is within a factor of 3 of the optimal GRP tour. This approach is especially relevant in geometric graphs, such as those in the Euclidean plane, where shortest connections and matchings can be computed efficiently.

References

- [1] Christos H. Papadimitriou. The euclidean travelling salesman problem is np-complete. *Theoretical Computer Science*, 4(3):237–244, 1977.
- [2] Jack Edmonds. Optimum branchings. *Journal of Research of the National Bureau of Standards*, October 1967.
- [3] P. M. Vaidya. Geometry helps in matching. *SIAM Journal on Computing*, 18(6):1201–1225, 1989.
- [4] K.R. Varadarajan. A divide-and-conquer algorithm for min-cost perfect matching in the plane. In *Proceedings of the 39th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 320–329, 1998.
- [5] Otakar Borůvka. O jistém problému minimálním. *Práce Moravské přírodovědecké společnosti*, 3(3):37–58, 1926.
- [6] Joseph B. Kruskal. On the shortest spanning subtree of a graph and the traveling salesman problem. *Proceedings of the American Mathematical Society*, 7(1):48–50, 1956.
- [7] R. C. Prim. Shortest connection networks and some generalizations. *The Bell System Technical Journal*, 36(6):1389–1401, 1957.